ISQua 43rd International Conference · Vancouver, Canada (19–22 Sept 2027) | Quality & Safety in Complex Systems

Co-Producing a SEIPS & Physical Poka-Yoke Quality Control Circle Bundle to Cut Bronchoalveolar Lavage Hypoxemia

Multidisciplinary QI Team | Department of Critical Care & Specialized Nursing, Fictional Tertiary Medical Center (Sample — Fictional Data)

SAMPLE
FICTIONAL

63.3% Relative Reduction in Intra-Procedure Hypoxemia (38.46% -> 14.10%, p < 0.001) While Saving 3.8 Min/Case Nursing Time

PRIMARY OUTCOME
38.46% -> 14.10% (n=319)

PERLA RUN-CHART
6-Week Shift < Median (p<0.05)

BALANCING MEASURE
9.0 -> 5.2 Min (-42% Nurse Load)

OPEN-ACCESS TOOLKIT
3D Poka-Yoke CAD & WMTY Card

1. INTRODUCTION & WAI-vs-WAD SYSTEM GAP

- **Clinical Burden:** Geriatric RICU patients undergoing bronchoalveolar lavage (BAL) exhibited a 38.46% baseline hypoxemia rate (60/156), prolonging ventilation and ICU stay.
- **WAI vs WAD Friction:** Written SOPs assumed unhurried two-nurse verification (Work-as-Imagined), whereas heavy night shifts forced manual estimation without angle/thermal locks (Work-as-Done; echoing ISQua SES-3944 ~3s SOP click-throughs).
- **Policy Alignment:** Directly targets China's 2026 National Patient Safety Goal (NIT-2026-IV) & WHO Global Patient Safety Action Plan.
- **Study Objectives:** Reduce intra-BAL hypoxemia below 17.60% within 6 months via a SEIPS 3.0 & Poka-Yoke QCC bundle without increasing nursing workload.

2. GEMBA DIAGNOSIS & HYPOTHESIS FALSIFICATION

Candidate Cause (SEIPS)	Gemba Audit (n=156)	Defect %	Verdict
No 3-node risk hard-stop	EHR log & bedside audit	35.0%	True Cause
Manual lavage temp/angle drift	IR sensor & angle gauge	28.3%	True Cause
Passive verbal pre-op coaching	Patient recall check	18.4%	True Cause
Ward ambient temp fluctuation	30-day dual-shift log	3.3%	Falsified
Catheter stockout in supply	90-day HIS inventory	0.0%	Falsified

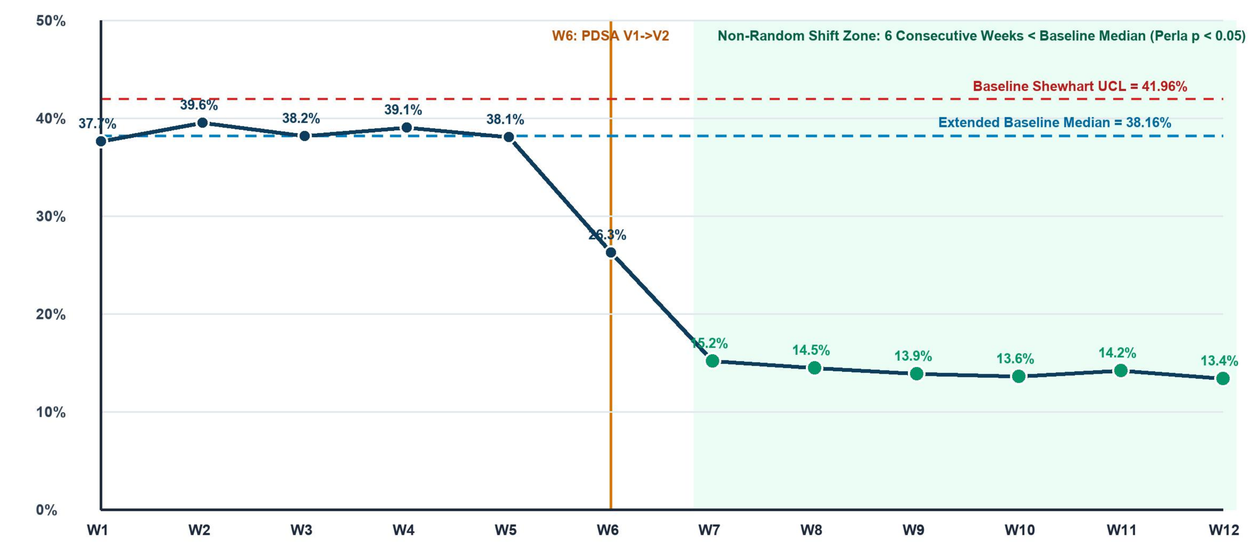
3. STRONG-ACTION PDSA RAMPS (VA RCA2 HIERARCHY)

- **[STRONG] PDSA 1 · EHR 3-Point Risk Hard-Stop:** Embedded mandatory pre/intra/post SpO2 & frailty scoring into PDA; blocks scope sign-out if high-risk pre-oxygenation protocol is unconfirmed.
- **[STRONG] PDSA 2 · V1->V2 Physical Poka-Yoke:** V1 clamp required 2 nurses and slowed setup -> redesigned V2 snap-fit 37C constant-temp sleeve & 30-45 deg airway port lock (30s single-nurse operation).
- **[CO-DESIGN] PDSA 3 · WMTY Patient Partnership:** Co-designed a 90-second visual breathing guide & bedside 'What Matters to You?' card with 5 recovered patients (386 verified QR scans).
- **Safety-II Positive Deviance:** Standardized secret suction-rhythm cues observed from zero-hypoxemia senior nurse shifts across all 28 ward nurses.

4. TIME-SERIES RESULTS: 12-WEEK PERLA RUN CHART & SPC

Weekly Procedure-Related Hypoxemia Rate (%) — 12-Week Run Chart & Shewhart Limits

Baseline Median: 38.16% (W1-W5, n=156) | PDSA V1->V2 Ramp: W6 | Sustained Strong-Action Mean: 14.13% (W7-W12, n=163, p<0.001)



Standards: NHS Making Data Count & Perla RJ et al. (BMJ Qual Saf 2011;20:46-51). Special-cause improvement confirmed; nursing procedure time simultaneously dropped 9.0 -> 5.2 min.

- **Statistical & SPC Proof:** Primary hypoxemia fell from 38.46% (60/156) to 14.10% (23/163; chi-square = 24.61, p < 0.001), with 6 consecutive post-ramp weeks below the 38.16% baseline median.

5. BALANCING MEASURES, PREMs & GREEN-CARE IMPACT

Domain	Outcome / Balancing Metric	Baseline	Post-QCC	Effect (p-value)
Primary Safety	Intra-BAL Hypoxemia Rate	38.46%	14.10%	-63.3% (p<0.001)
Balancing #1	Nursing Time / Case (Workload)	9.0 min	5.2 min	-3.8 min (p<0.01)
Balancing #2	Unplanned Extubation / Injury	1.28%	0.00%	Zero Harm
Co-Production	Validated Patient PREMs (0-100)	81.2 pts	96.4 pts	+15.2 (p<0.001)
ISQua Green	Rework Consumable Carbon	1.42 kg	0.61 kg	-57% CO2e

6. SUSTAINABILITY, HOSPITAL SPREAD & GOVERNANCE

- **Institutionalization:** Codified into Hospital Standard SOP-RICU-2026-08 with quarterly 3-month post-audit sustaining >96.2% bundle fidelity across 3 ICU units.
- **Scalability:** 3D-printed poka-yoke sleeve (< USD 12/unit) + EHR rule requires zero extra staffing, enabling rapid transfer to resource-constrained hospitals.
- **Ethics & COI:** Sample congress poster layout — all institution details, audit counts, and clinical outcomes are fictional.